
KUET Kinship

Software Documentation — GitHub README Edition (Updated for v1.0.3-stable)

Document purpose: updated documentation reflecting the KUET Kinship v1.0.3-stable release — covering the new Profile system, the gamified contribution & recognition model, the individual Campus Toolkit & Settings features, the KUET-green design language, the UI/UX philosophy, the populated Rooms & Labs data, the populated Lecture Notes links, and the resolved course-year filtering issue.

1. Project Overview

KUET Kinship is an Android application designed to help KUET students access academic and campus-related information, and to connect with their campus community, from one place. The implemented scope centers on course information, examination information, lecture notes, rooms/labs with navigation support, and — as of v1.0.3-stable — a student Profile system with a gamified contribution and recognition model.

The application now presents itself with a social-media-style profile experience: each verified KUETian has an identity card (name, roll number, hall, department and batch), a contribution level and point total, earned achievement badges, and a personal toolkit of campus-related settings and saved resources.

The current release is best described as a lightweight, locally packaged academic and campus information application, now paired with a community-recognition layer, rather than a fully server-backed, synchronized platform.

2. Project Goals

- Provide students with organized academic information.
- Make course and examination information easier to access.
- Provide a structured place for lecture notes.
- Help students find rooms and laboratories more easily, using real faculty room and contact data.
- Give every KUETian a verified, social-media-style profile identity within the app.
- Encourage active contribution to the platform through a gamified points, level, and achievement system.
- Keep the application simple and convenient for everyday student use.
- Present a clean, elegant, and intuitive user interface built around KUET's own branding.
- Keep data access and processing lightweight enough for the current release.

3. Current Feature Scope

Feature	Status	Description
Course Selection	Implemented	Year → Term → Course selection and course information.
Examination	Implemented	Exam-related information and details.

Feature	Status	Description
Lecture Notes	Implemented (updated)	Access to lecture notes; real, working links to the actual note materials have been added for the covered courses.
Rooms & Labs	Implemented (updated)	Room/lab information with navigation support; real faculty phone numbers and physical room locations have been added.
Profile	Implemented (new)	Social-media-style student profile: verified identity, bio, department/batch, contribution level and points, and achievement badges.
Gamification & Achievements	Implemented (new)	Points, levels, tier progression, and unlockable achievement badges awarded for contribution activity.
Campus Toolkit & Settings	Implemented (new)	A set of individual profile-linked features: shared guides/notes, saved offline resources, hall & mess preferences, settings & privacy, and help & campus emergency contacts.
Updates	Not implemented	Reserved for future development.
Settings (system-level)	Not implemented	Reserved for future development; a Settings & Privacy entry point exists in the Profile toolkit, but is a placeholder.
Real Database	Not implemented	Persistent/server-backed database was not included in this release.
Search Algorithm	Not implemented	Dedicated search functionality was not included in this release.
Online Sync	Not implemented	No online synchronization layer is included in the current scope.
Data Validation Mechanism	Not implemented	A dedicated validation layer is not included in the current scope.

4. High-Level Architecture

The application follows an Activity-and-Fragment based Android architecture. MainActivity acts as the main entry point, HomeFragment provides the primary dashboard/navigation area, and feature-specific Activities/Fragments handle the individual modules. The v1.0.3-stable release adds a Profile-oriented branch to this structure.

- **MainActivity** — initializes the main application structure and navigation.
- **HomeFragment** — displays the main dashboard and entry points to major features.
- **CourseSelectionFragment** — handles course-selection flow.
- **YearSelectionFragment** — allows selection of academic year.
- **TermSelectionFragment** — allows selection of academic term.

- **ExamAndStuffActivity** — provides examination-related information.
- **LectureNotesActivity** — provides access to lecture notes, now backed by real note links.
- **RoomsAndLabsActivity** — displays room and laboratory information, now including real faculty numbers and room locations.
- **RoomNavigateActivity** — supports room navigation.
- **ProfileFragment** — renders the student identity card, level/points summary, tier progression, and achievement badges, and hosts the Campus Toolkit & Settings entry points (new in v1.0.3-stable).
- **Bottom navigation (Home / Feed / Profile)** — the primary navigation surface, with Profile as a first-class destination alongside Home and Feed.

The APK/documentation evidence supports AndroidX-related components, generated View/Data Binding classes, and Kotlin coroutine-related components. Exact backend/API/database behavior should not be inferred beyond what the APK and supplied documentation establish.

5. Application Flow

Academic flow: Application launch → MainActivity → HomeFragment → Course Selection → Year Selection → Term Selection → Course information.

Examination flow: Home → Exam section → relevant examination information → details.

Learning flow: Home → Lecture Notes → relevant academic/course information → lecture note (now opening a real, working link).

Campus flow: Home → Rooms & Labs → room/lab information (now with real faculty numbers and locations) → navigation when available.

Identity & community flow (new): Bottom navigation → Profile → identity card, level/points, and achievement badges → Campus Toolkit & Settings → individual toolkit feature.

6. Detailed User Flow

Course

- Open the app.
- Open Course Selection.
- Choose your year.
- Choose your term.
- Choose a course.
- View the available course information.

Examination

- Open the app.
- Open the Exam section.

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- Select the relevant examination information.
 - Open and view the details.

Lecture Notes

- Open the app.
- Open Lecture Notes.
- Select the relevant academic/course information.
- Open and read the note via its real, linked source.

Rooms and Labs

- Open the app.
- Open Rooms & Labs.
- Select the required room or laboratory.
- View the room location and the relevant faculty contact number.
- Use navigation when available.

Profile (new)

- Open the app and tap Profile in the bottom navigation.
- View the verified identity card: name, roll number, hall, and Dept./batch tag.
- Review current level, point total, tier-progression bar, and earned achievement badges.
- Open Campus Toolkit & Settings to reach an individual feature (Shared Guides & Notes, Saved Offline Resources, Hall & Mess Preferences, Settings & Privacy, or Help & Campus Emergency).
- Optionally edit bio/info, view the Digital ID, or sign out.

7. User Profile & Identity

v1.0.3-stable introduces a dedicated Profile section, modeled after a social-media profile page so that it feels familiar and personal rather than purely administrative. Each profile is tied to a verified KUETian identity rather than an anonymous account.

Identity card

- **Verification:** profiles carry a “Verified” badge, confirming that the account belongs to an authenticated KUET student.
- **Core identity fields:** full name, KUET roll number, and residential hall.
- **Academic tag:** department and batch/year (e.g. “Dept. of CSE · Batch 2k25 (1st Year)”).
- **Bio:** a short, student-editable description of interests and how the student contributes to the community.
- **Profile photo** and a share action, so a profile (or Digital ID) can be shared or shown when needed.
- **Digital ID:** a quick-access digital identity card derived from the profile.

The identity card is intentionally the first thing shown on the Profile screen, reinforcing that Kinship is a verified-KUETian space before any gamified elements are shown.

8. Gamification & Contribution System

To encourage students to keep contributing content and helping juniors/peers, the Profile section layers a points-based gamification model on top of the identity card.

Points, levels, and tiers

- Every contribution a student makes toward the community (sharing guides and notes, uploading lab reports, helping with course/portal information, and similar activity) earns points.
- Accumulated points determine the student's current **Level** (e.g. Level 4) and an associated **title** (e.g. "Campus Senior Mentor"), which is displayed prominently on the profile.
- A **tier-progression bar** shows how many points remain before the next tier (e.g. "80 pts to Campus Guide Elite (L5)"), giving a clear, visual contribution goal.
- A relative-standing badge (e.g. "Top 5% Batch") benchmarks a student's contribution against peers in the same batch, adding a light competitive/social incentive.

Achievement badges

Distinct, specific achievements are awarded for meaningful contribution milestones rather than a single generic point counter. Examples surfaced on the profile include:

- **Course-specific recognition** (e.g. "CSE 301 Wayfinder") for helping other students navigate a particular course.
- **Usage-based recognition** (e.g. "Dining Portal Whiz (150+)") for heavy, helpful engagement with a specific campus utility.
- **Content-contribution recognition** (e.g. "12 Lab Reports Shared") for material shared with the community.

This points → level → title → badge structure is designed to make contribution visible and rewarding, turning helpful behavior (sharing notes, answering questions, keeping information up to date) into recognizable in-app status.

9. Campus Toolkit & Settings

The Profile screen hosts a "Campus Toolkit & Settings" panel. Each row in this panel is its own individual feature with a dedicated screen, summarized below.

My Shared Guides & Notes

A personal library of everything the student has contributed — uploaded class suggestions and lab-report shares — with an activity indicator for unread updates or reactions on shared items.

Saved Offline Resources

A local, offline-accessible store of frequently needed campus references: syllabus copies, hall/campus floorplans, and bus-line information, available even without a network connection.

Hall & Mess Preferences

Residential settings for the student's assigned hall and room, together with dining/mess preferences such as an auto-meal toggle, shown with an "Active" status indicator.

Settings & Privacy

Account-level controls covering push-notification preferences, dark theme, and two-factor authentication.

Help & Campus Emergency

A quick-access directory for urgent or administrative contacts: the proctorial cell, the medical center, and the IT help desk.

10. Design Language & Branding

KUET Kinship's visual identity is built around KUET's own institutional color. Green was chosen as the primary branding color specifically because it is KUET's own color, so that the app visually reads as belonging to the KUET community rather than as a generic, unaffiliated tool.

- **Primary color — KUET Green:** used for the bottom navigation bar, primary buttons, section headers, active/status indicators (e.g. the "Active" and "Verified" tags), and the tier-progression bar, so the strongest brand color is tied to the app's most important actions and confirmations.
- **Soft neutral backgrounds:** light, low-saturation backgrounds are used behind cards and panels so that the green branding and the student's own content (photos, badges, text) remain the visual focus.
- **Accent colors:** warm accent tones (amber/yellow for level and star indicators, soft red for the Sign Out action) are used sparingly to draw attention to specific, meaningful actions without competing with the primary green identity.
- **Consistency:** the same green identity is carried across navigation, cards, badges, and buttons, so the branding stays recognizable and consistent throughout every screen of the app.

11. UI/UX Design Philosophy

Alongside the visual branding, the interface is deliberately kept clean, elegant, and intuitive, and this is treated as a feature of the app in its own right rather than an incidental side effect.

- **Clean layout:** information is grouped into clearly separated cards (identity, level/achievements, toolkit) so each screen stays easy to scan rather than dense or cluttered.
- **Elegant, minimal styling:** rounded cards, generous spacing, and a restrained color palette are used instead of heavy borders or visual noise.
- **Intuitive navigation:** a persistent three-tab bottom navigation (Home, Feed, Profile) keeps the app's main areas one tap away at all times.
- **Familiar, low-friction patterns:** the Profile screen mirrors patterns students already know from social-media apps (identity card, stats, settings list), reducing the learning curve for new users.
- **Clear status communication:** badges and tags (Verified, Active, Top 5% Batch, notification dots) communicate state at a glance without requiring the student to open a sub-screen.

12. Technology and Implementation

- Platform: Android.
- Architecture style: Activity and Fragment based Android application.
- UI implementation includes generated View/Data Binding classes.
- AndroidX-related components are present.
- Kotlin coroutine-related components are present in the application package.
- The release referenced for the base architecture review is kuet_kinship_ver_1.0.2-stable.apk (archive size at analysis time: 14,193,931 bytes; SHA-256: 3659e0d425cd42e5..., included only as a build-identification aid); the current documented release is v1.0.3-stable (Build 103), which adds the Profile, gamification, and Campus Toolkit & Settings features described above.
- Product credit: “Crafted with ❤ by KUET TITANIUM for KUETians.”

13. Data and Backend Design

The current release does not implement a real database, online synchronization, or a dedicated data-validation mechanism. The application should therefore be treated as using packaged/static or otherwise locally handled information for the currently implemented modules, including the newly added Rooms & Labs contact/location data and Lecture Notes links. The exact storage representation should be confirmed from the source repository if a source-level architecture description is required.

This design keeps the current build comparatively simple and avoids the additional processing, networking, synchronization, indexing, and validation logic that would be required by a larger data-driven system. The points/levels/achievements shown on the Profile screen should likewise be understood as running on this same lightweight, locally handled data model unless a source-level review establishes otherwise.

14. Performance / Time-Complexity Constraints

Several advanced data-management features were intentionally left out of the current release because implementing them would increase algorithmic and system complexity. In particular:

- Real database integration would require database schema design, queries, persistence management, indexing, and synchronization considerations.
- A dedicated search algorithm would add search/indexing logic and additional processing over the available information.
- Online synchronization would introduce network requests, conflict handling, caching, retry logic, and consistency management — this also applies to keeping points/levels/achievements synchronized across devices in the future.
- A formal data-validation mechanism would require validation rules, error handling, and processing at data-entry/loading boundaries.

15. Known Issues

Course List Year Filtering — Resolved in v1.0.3-stable

Previous behavior: selecting 1st, 2nd, 3rd, or 4th year could result in the same course list being displayed, instead of each year showing only its relevant courses.

Status: fixed. The selected year is now correctly captured, passed to the course-list logic, and used to filter the course dataset, so each academic year displays its own course list.

Recommended follow-up: repeat regression testing across all four academic years after any future change to the course-selection logic, to confirm the fix continues to hold.

16. Limitations

- Updates and system-level Settings are not implemented.
- A real database is not implemented in the current release.
- A dedicated search algorithm is not implemented.
- Online data synchronization is not implemented (this also applies to the Profile/points/achievements data, which is not yet synchronized across devices).
- A dedicated data-validation mechanism is not implemented.
- The exact server-side/database architecture cannot be fully established from the APK alone.
- Feature availability (including Rooms & Labs data and Lecture Notes links) may depend on the data included in the current build.
- Without online sync, content can become outdated unless a new application build or another supported update mechanism is provided.

17. Future Improvements

- Implement Updates and system-level Settings modules.
- Add a real database for structured and persistent data management, including Profile, points, and achievement data.
- Implement efficient search and, where appropriate, indexing.
- Add online synchronization with a suitable backend/API, so contribution points and achievements sync across devices.
- Introduce a dedicated data-validation mechanism.
- Expand the gamification system with additional badge categories and possibly a batch/hall leaderboard.
- Improve error handling and empty-state messages.
- Add automated tests for year, term, course selection, and for the Profile/points/achievements logic.
- Document APIs, data sources, and database structure.
- Add continuous integration and release automation to the GitHub workflow.

18. Testing and QA

- Test every implemented feature on supported Android versions.
- Test navigation between Activities and Fragments, including the Home / Feed / Profile bottom navigation.
- Verify year and term selections independently, and confirm the course-year filtering fix holds for all four years.
- Verify lecture-note links open correctly and examination content displays correctly.
- Verify room/lab information, including the newly added faculty numbers and room locations, and navigation behavior.
- Verify Profile data: identity fields, verification badge, points/level display, tier-progression bar, and achievement badges.
- Verify each Campus Toolkit & Settings entry (Shared Guides & Notes, Saved Offline Resources, Hall & Mess Preferences, Settings & Privacy, Help & Campus Emergency) opens and displays correctly.
- Test behavior when data is missing or unavailable.
- Record reproducible bugs with steps, expected result, actual result, severity, and evidence.
- For future database/search/sync/validation features, add dedicated performance and correctness tests.

19. Release and Documentation Status

This document is an updated GitHub/README-oriented software documentation reference for KUET Kinship v1.0.3-stable (Build 103). It distinguishes implemented features from intentionally unimplemented features, documents the new Profile, gamification, Campus Toolkit & Settings, design-language, and UI/UX sections, records the resolution of the course-year filtering issue, and records the current limitations and future development direction.

20. Contribution Guidelines

- Create a separate branch for each feature or bug fix.
- Use clear commit messages.
- Keep changes focused and reviewable.
- Test changes before creating a pull request.
- For bugs, include clear reproduction steps and expected versus actual behavior.
- Update the documentation whenever a feature or architecture change is introduced.

21. Conclusion

KUET Kinship v1.0.3-stable provides a focused Android solution for accessing academic and campus information, now paired with a verified, social-media-style Profile and a gamified contribution and recognition system. The implemented scope includes Course Selection, Examination, Lecture Notes (with real links), Rooms & Labs (with real faculty numbers and locations), Profile, Gamification & Achievements, and the Campus Toolkit & Settings features. The release intentionally keeps the system lightweight by excluding a real

database, dedicated search algorithm, online synchronization, and a formal data-validation layer. The previously known course-year filtering issue has been resolved. These points should be considered when evaluating scalability and data freshness, while also providing a clear roadmap for future versions.

Source basis: This update was grounded in the supplied previous software documentation, the supplied v1.0.2-stable APK, and the supplied v1.0.3 (Build 103) Profile-screen screenshots. Where the underlying source cannot establish an implementation detail beyond what is directly observable, the documentation explicitly avoids presenting that detail as a confirmed fact.

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